

Advanced (Habanero)

- Q1.** What is the graph of the system $x + y > 2$ and $x - y < 1$?
- (A) Two intersecting lines
(B) Two parallel lines
(C) A single line
(D) No lines
(E) A point
- Q2.** The inequality $2x - 3y > 5$ can be graphed by!
- (A) Plotting two points
(B) Drawing a vertical line
(C) Drawing a horizontal line
(D) Shading above the line
(E) Finding the y -intercept
- Q3.** What does the region $x > 0$ and $y > 0$ represent?
- (A) First quadrant
(B) Second quadrant
(C) Third quadrant
(D) Fourth quadrant
(E) Origin
- Q4.** Which of the following inequalities has the solution region $y < -x + 2$?
- (A) $y > x - 2$
(B) $y < x + 2$
(C) $y > -x - 2$
(D) $y < x - 2$
(E) $x > y + 2$
- Q5.** The inequality $y \leq 2x - 1$ is equivalent to!
- (A) $y > 2x + 1$
(B) $y < 2x + 1$
(C) $y \geq 2x + 1$
(D) $y \leq -2x - 1$
(E) $y \geq -2x + 1$
- Q6.** For the system $x + y > 3$ and $x - y > 1$, the solution set is!
- (A) The first quadrant
(B) The second quadrant
(C) The third quadrant
(D) The fourth quadrant
(E) The entire plane
- Q7.** When graphing $y > x^2 - 4$, what is the result?
- (A) A parabola that opens up
(B) A parabola that opens down
(C) A horizontal line
(D) A vertical line
(E) A circle
- Q8.** The system $x > 2$ and $y < 1$ has a solution region that is!
- (A) A triangle
(B) A rectangle
(C) A trapezoid
(D) A quarter circle
(E) A half plane

Open Ended Questions (FRQ)

- Q9.** Graph the system $x + y \leq 4$ and $x - y \geq -2$. Find the vertices of the solution region and provide the x - and y -intercepts.
- Q10.** The feasible region for a company's production is defined by $2x + 3y \leq 12$, $x + 2y \leq 8$, $x \geq 0$, and $y \geq 0$. Determine the vertices of this feasible region and calculate the maximum value of $3x + 4y$. Provide calculations and graph.

Chef's Tips

- Encourage students to check their work by plugging points into the inequalities.
- Caution students to avoid common pitfalls such as not switching the inequality when multiplying or dividing by a negative number.
- Emphasize the importance of labeling the x - and y -axes and providing a key for the graph.